

15.20 def

2026-06-01

15.20 This exercise uses data from the STAR experiment introduced to illustrate fixed and random effects for grouped data. In the STAR experiment, children were randomly assigned within schools into three types of classes: small classes with 13-17 students, regular-sized classes with 22-25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file star.

```
load("/Users/liaochenghao/Desktop/Financial Econometrics/Homework/HW9/Data/star.rdata")
df <- star
names(df)
```

```
## [1] "id"      "schid"    "tchid"    "tchexper"  "absent"
## [6] "readscore" "mathscore" "totalscore" "boy"      "white_asian"
## [11] "black"    "tchwhite"  "tchmasters" "freelunch" "schurban"
## [16] "schrural" "small"    "regular"   "aide"
```

- d. Reestimate the model in part (a) with school random effects. Compare the results with those from parts (a) and (b). Are there any variables in the equation that might be correlated with the school effects? Use the LM test for the presence of random effects.

```
# id = individual/student identifier, used as the "time" index here
pdata <- pdata.frame(df, index = c("schid", "id"))

# Part (a): pooled OLS
model_pool <- plm(
  readscore ~ small + aide + tchexper + boy + white_asian + freelunch,
  data = pdata,
  model = "pooling"
)
```

```
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
```

```
# Part (b): school fixed effects
model_fe <- plm(
  readscore ~ small + aide + tchexper + boy + white_asian + freelunch,
  data = pdata,
  model = "within",
  effect = "individual"
)
```

```
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
```

[illegible]

```
summary(model_re)
```

```
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
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```

```
## Oneway (individual) effect Random Effect Model
```

```
## (Swamy-Arora's transformation)
```

```
##
```

```
## Call:
```

```
## plm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
##      freelunch, data = pdata, effect = "individual", model = "random",
##      random.method = "swar")
```

```
##
```

```
## Unbalanced Panel: n = 79, T = 34-137, N = 5766
```

```
##
```

```
## Effects:
```

```
##               var std.dev share
```

```
## idiosyncratic 751.43   27.41 0.829
```

```
## individual    155.31   12.46 0.171
```

```
## theta:
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```

```
## 0.6470 0.7225 0.7523 0.7541 0.7831 0.8153
```

```
##
```

```
## Residuals:
```

```
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```

```
## -97.4830 -17.2361 -3.2822  0.0372 12.8027 192.3460
```

```
##
```

```
## Coefficients:
```

```
##               Estimate Std. Error z-value Pr(>|z|)
```

```
## (Intercept) 436.126774   2.064782 211.2217 < 2.2e-16 ***
```

```
## small       6.458722   0.912548  7.0777 1.466e-12 ***
```

```
## aide       0.992146   0.881159  1.1260 0.2602
```

```
## tchexper    0.302679   0.070292  4.3060 1.662e-05 ***
```

```
## boy       -5.512081   0.727639 -7.5753 3.583e-14 ***
```

```
## white_asian 7.350477   1.431376  5.1353 2.818e-07 ***
```

```
## freelunch -14.584332   0.874676 -16.6740 < 2.2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.

## Total Sum of Squares:      6158000

## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.

## Residual Sum of Squares: 4332100
## R-Squared:      0.29655
## Adj. R-Squared: 0.29582
## Chisq: 493.205 on 6 DF, p-value: < 2.22e-16
```

```
plmtest(
  model_pool,
  effect = "individual",
  type = "bp"
)
```

```
## Warning in index.pindex(x = anindex, which = which): an index variable not
## being the individual index is called 'id'. Likely, any results are distorted.

##
## Lagrange Multiplier Test - (Breusch-Pagan)
##
## data:  readscore ~ small + aide + tchexper + boy + white_asian + freelunch
## chisq = 6677.4, df = 1, p-value < 2.2e-16
## alternative hypothesis: significant effects
```

I reestimate the model using school random effects. The estimated random effects equation is

$$READSCORE_{hat} = 436.1268 + 6.4587SMALL + 0.9921AIDE + 0.3027TCHEXPER - 5.5121BOY + 7.3505WHITE_ASIAN \quad (1)$$

Compared with the pooled OLS results in part (a), the coefficient on *SMALL* increases from 5.8228 to 6.4587 and remains statistically significant. It is also very close to the fixed effects estimate of 6.4902. Therefore, the conclusion that students in small classes perform better in reading does not change.

The coefficient on *AIDE* is 0.9921 and remains statistically insignificant. Thus, as in parts (a) and (b), there is no evidence that a teacher aide significantly improves reading scores.

The coefficient on *TCHEXPER* is 0.3027 and is statistically significant. This is smaller than the pooled OLS estimate of 0.4925, but close to the fixed effects estimate of 0.2856. Thus, students taught by more experienced teachers still tend to score higher on reading tests, although the estimated effect is smaller after accounting for school-level effects.

The coefficient on *BOY* remains negative and statistically significant, indicating that boys score lower than girls. The coefficient on *WHITE_ASIAN* remains positive and statistically significant, and the coefficient on *FREELUNCH* remains negative and highly significant. These conclusions are similar to those from the pooled OLS and fixed effects models.

The random effects model assumes that the school-specific effect is uncorrelated with all explanatory variables. This assumption may be questionable for variables such as *WHITE_ASIAN* and *FREELUNCH*, because schools may differ in racial composition, neighborhood income, and socioeconomic background. *TCHEXPER* may also be correlated with school effects if more experienced teachers tend to work in certain schools. In contrast, *SMALL* and *AIDE* are less likely to be correlated with school effects because students were randomly assigned within schools to different class types in the STAR experiment.

The Breusch-Pagan LM test for random effects gives a chi-square statistic of 6677.4 with a p-value less than 2.2e-16. Therefore, I reject the null hypothesis that the variance of the school random effect is zero. This provides strong evidence for the presence of school random effects.

Overall, the LM test suggests that random effects are present and that pooled OLS is not adequate. However, if some regressors are correlated with the school effects, the random effects estimator may be inconsistent, and the fixed effects estimator would be more reliable.

-
- e. Using the t-test statistic in equation (15.36) and a 5% significance level, test whether there are any significant differences between the fixed effects and random effects estimates of the coefficients on *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, and *FREELUNCH*. What are the implications of the test outcomes? What happens if we apply the test to the fixed and random effects estimates of the coefficient on *BOY*?

$$t = \frac{b_{FE,k} - b_{RE,k}}{[\hat{v}ar(b_{FE,k}) - \hat{v}ar(b_{RE,k})]^{1/2}} = \frac{b_{FE,k} - b_{RE,k}}{[se(b_{FE,k})^2 - se(b_{RE,k})^2]^{1/2}} \quad (15.36)$$

```
# Extract FE and RE coefficient estimates
b_fe <- coef(model_fe)
b_re <- coef(model_re)

# Extract standard errors
se_fe <- sqrt(diag(vcov(model_fe)))
se_re <- sqrt(diag(vcov(model_re)))

# Variables required in the question
vars <- c("small", "aide", "tchexper", "white_asian", "freelunch")

# Hausman t-test statistic from equation (15.36)
hausman_t <- (b_fe[vars] - b_re[vars]) / sqrt(se_fe[vars]^2 - se_re[vars]^2)

# Two-sided p-values
p_values <- 2 * (1 - pnorm(abs(hausman_t)))

# Put results into table
hausman_table <- data.frame(
  Variable = vars,
  FE_Estimate = b_fe[vars],
  RE_Estimate = b_re[vars],
  FE_SE = se_fe[vars],
  RE_SE = se_re[vars],
  t_stat = hausman_t,
  p_value = p_values
)

hausman_table
```

	Variable	FE_Estimate	RE_Estimate	FE_SE	RE_SE
##	small	6.4902305	6.4587216	0.91296175	0.91254764
##	aide	0.9960875	0.9921460	0.88169306	0.88115884
##	tchexper	0.2855668	0.3026787	0.07084451	0.07029195
##	white_asian	8.0280192	7.3504772	1.53565617	1.43137578
##	freelunch	-14.5935724	-14.5843317	0.88000649	0.87467623
##	t_stat	p_value			
##	small	1.14600764	0.2517920		
##	aide	0.12843803	0.8978023		
##	tchexper	-1.93771666	0.0526578		
##	white_asian	1.21807432	0.2231957		
##	freelunch	-0.09555102	0.9238772		

Using the t-test statistic in equation (15.36), I compare the fixed effects and random effects estimates for *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, and *FREELUNCH*.

For *SMALL*, the test statistic is 1.1460 with a p-value of 0.2518. For *AIDE*, the test statistic is 0.1284 with a p-value of 0.8978. For *TCHEXPER*, the test statistic is -1.9377 with a p-value of 0.0527. For *WHITE_ASIAN*, the test statistic is 1.2181 with a p-value of 0.2232. For *FREELUNCH*, the test statistic is -0.0955 with a p-value of 0.9239.

At the 5% significance level, none of these test statistics is statistically significant. Therefore, I fail to reject the null hypothesis that the fixed effects and random effects estimates are the same for these coefficients.

This implies that there is no strong evidence that *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, or *FREELUNCH* is correlated with the school-specific unobserved heterogeneity. Therefore, based on these coefficient-by-coefficient Hausman tests, the random effects estimates do not appear to be significantly different from the fixed effects estimates for these variables.

The coefficient on *TCHEXPER* is close to being significant, since its p-value is 0.0527, but it is still slightly above 0.05. Thus, at the 5% level, it is not significant.

If the same test is applied to *BOY*, the test statistic cannot be computed properly. The reason is that the estimated standard error of the fixed effects estimator is slightly smaller than that of the random effects estimator for *BOY*. Therefore, $se(b_{FE})^2 - se(b_{RE})^2$ is negative, and the denominator of the t-statistic involves the square root of a negative number. As a result, the t-statistic is undefined for *BOY*.

-
- f. Create school-averages of the variables and carry out the Mundlak test for correlation between them and the unobserved heterogeneity.

```
df$student_id <- df$id

# Keep only complete observations used in the model
df2 <- df[complete.cases(df[, c(
  "readscore", "small", "aide", "tchexper",
  "boy", "white_asian", "freelunch", "schid", "student_id"
)]), ]

# Create school averages
df2$mean_small <- ave(df2$small, df2$schid, FUN = mean)
df2$mean_aide <- ave(df2$aide, df2$schid, FUN = mean)
df2$mean_tchexper <- ave(df2$tchexper, df2$schid, FUN = mean)
df2$mean_boy <- ave(df2$boy, df2$schid, FUN = mean)
```

```

df2$mean_white_asian <- ave(df2$white_asian, df2$schid, FUN = mean)
df2$mean_freelunch <- ave(df2$freelunch, df2$schid, FUN = mean)
pdata2 <- pdata.frame(df2, index = c("schid", "student_id"))

model_mundlak <- plm(
  readscore ~ small + aide + tchexper + boy + white_asian + freelunch +
    mean_small + mean_aide + mean_tchexper + mean_boy +
    mean_white_asian + mean_freelunch,
  data = pdata2,
  model = "random",
  effect = "individual",
  random.method = "walhus"
)

summary(model_mundlak)

## Oneway (individual) effect Random Effect Model
## (Wallace-Hussain's transformation)
##
## Call:
## plm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
##     freelunch + mean_small + mean_aide + mean_tchexper + mean_boy +
##     mean_white_asian + mean_freelunch, data = pdata2, effect = "individual",
##     model = "random", random.method = "walhus")
##
## Unbalanced Panel: n = 79, T = 34-137, N = 5766
##
## Effects:
##               var std.dev share
## idiosyncratic 751.43   27.41 0.829
## individual    155.31   12.46 0.171
## theta:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6470 0.7225 0.7523 0.7541 0.7831 0.8153
##
## Residuals:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -98.9821 -16.9749  -3.1524   0.0401  12.8252  193.2527
##
## Coefficients:
##               Estimate Std. Error z-value Pr(>|z|)
## (Intercept)   460.053192  19.569578  23.5086 < 2.2e-16 ***
## small         6.490231   0.912839   7.1099 1.161e-12 ***
## aide          0.996087   0.881574   1.1299 0.25852
## tchexper      0.285567   0.070835   4.0314 5.544e-05 ***
## boy          -5.455941   0.727491  -7.4997 6.398e-14 ***
## white_asian    8.028019   1.535449   5.2285 1.709e-07 ***
## freelunch     -14.593572   0.879888 -16.5857 < 2.2e-16 ***
## mean_small    -18.987867  21.155710  -0.8975 0.36944
## mean_aide      16.556088  19.961335   0.8294 0.40687
## mean_tchexper   0.997469   0.595544   1.6749 0.09396 .
## mean_boy      -53.305170  24.222414  -2.2007 0.02776 *
## mean_white_asian -6.686636   6.037098  -1.1076 0.26804

```

```
## mean_freelunch    -3.601296    8.365600   -0.4305    0.66684
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    6158000
## Residual Sum of Squares: 4321800
## R-Squared:    0.29823
## Adj. R-Squared: 0.29676
## Chisq: 507.574 on 12 DF, p-value: < 2.22e-16
```

```
linearHypothesis(
  model_mundlak,
  c(
    "mean_small = 0",
    "mean_aide = 0",
    "mean_tchexper = 0",
    "mean_boy = 0",
    "mean_white_asian = 0",
    "mean_freelunch = 0"
  )
)
```

```
##
## Linear hypothesis test:
## mean_small = 0
## mean_aide = 0
## mean_tchexper = 0
## mean_boy = 0
## mean_white_asian = 0
## mean_freelunch = 0
##
## Model 1: restricted model
## Model 2: readscore ~ small + aide + tchexper + boy + white_asian + freelunch +
##      mean_small + mean_aide + mean_tchexper + mean_boy + mean_white_asian +
##      mean_freelunch
##
##   Res.Df Df    Chisq Pr(>Chisq)
## 1     5759
## 2     5753  6 13.709   0.03306 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

To carry out the Mundlak test, I first create school-level averages of SMALL, AIDE, TCHEXPER, BOY, WHITE_ASIAN, and FREELUNCH. I then include these school averages in the random effects model.

The Mundlak model is

$$READSCORE_{ij} = \beta X_{ij} + \gamma \bar{X}_j + \alpha_j + u_{ij}, \quad (2)$$

where $\bar{X}_j$ denotes the school-level averages of the explanatory variables.

The null hypothesis is that all coefficients on the school averages are jointly equal to zero:

$$H0: \gamma_{mean_small} = \gamma_{mean_aide} = \gamma_{mean_tchexper} = \gamma_{mean_boy} = \gamma_{mean_white_asian} = \gamma_{mean_freelunch} = 0. \quad (3)$$

The alternative hypothesis is that at least one of these school-average variables is statistically significant, implying correlation between the regressors and the unobserved school-level heterogeneity.

The joint Mundlak test gives a chi-square statistic of 13.709 with a p-value of 0.03306. Since the p-value is less than 0.05, I reject the null hypothesis at the 5% significance level.

Therefore, the school averages are jointly significant. This provides evidence that the regressors are correlated with the unobserved school effects. Hence, the key random effects assumption that the regressors are uncorrelated with the school-specific unobserved heterogeneity is rejected.

Looking at the individual coefficients, *MEAN_BOY* is statistically significant at the 5% level, and *MEAN_TCHEXPER* is marginally significant at the 10% level. The other school-average variables are not individually significant.

Overall, the Mundlak test suggests that the standard random effects estimator may be inconsistent. Therefore, the school fixed effects model is more reliable than the standard random effects model in this case.